

Mark schemes

Q1.

(a) $\mathbf{r} = 5\mathbf{j}$

Correct vector

B1

1

(b) $4t - 0.01t^2 = 0$

Equation based on i component

M1

Correct equation

A1

$t = 0$ or $t = 400$

Solving the quadratic

M1

$t = 400$

Selecting $t = 400$

A1

4

(c) $\mathbf{v} = (4 - 0.02t)\mathbf{i} + (-3 - 0.08t)\mathbf{j}$

Use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$

M1

Correct velocity vector

A1

$4 - 0.02t = 3 + 0.08t$

Equation using both components

M1

Correct equation

A1

$t = 10$

Correct t

A1

5

[10]

Q2.

(a) $19\mathbf{i} + 13\mathbf{j} = 35\mathbf{i} + 45\mathbf{j} + 8\mathbf{a}$

Constant acceleration equation to find a

M1

$$\mathbf{a} = \frac{19-35}{8}\mathbf{i} + \frac{13-45}{8}\mathbf{j} = 2\mathbf{i} - 4\mathbf{j}$$

Correct equation

A1

ag *Correct a from correct working*

A1

3

(b) $\mathbf{r} = (35\mathbf{i} + 45\mathbf{j})t + \frac{1}{2}(-2\mathbf{i} - 4\mathbf{j})t^2$

Use of constant acceleration equation

M1

Correct i component

A1

Correct j component

A1

3

(c) $\mathbf{r} = (3.5t - t^2)\mathbf{i} + (45t - 2t^2)\mathbf{j}$

Splitting into components correctly

B1

$$35t - t^2 = 300$$

Forming equation for one component

M1

$$t^2 - 35t + 300 = 0$$

Solving quadratic

M1

$$t = 15 \text{ or } t = 20$$

Two correct solutions

A1

$$4t - 2t^2 = 225$$

Forming correct second quadratic

B1

$$2t^2 - 45t + 225 = 0$$

$$t = 7.5 \text{ or } t = 15$$

Solving quadratic for two solutions

M1

$$t = 15 \text{ seconds}$$

Correct final solution

A1

7

[13]

Q3.

(a) $2\mathbf{i} + 14\mathbf{j} = \mathbf{u} + 2(2\mathbf{i} - 3\mathbf{j})$

Using constant acceleration equation to find $\mathbf{u}$
Correct equation

M1

Correct $\mathbf{u}$ from correct working

A1

$$\mathbf{u} = (2 - 4)\mathbf{i} + (14 + 6)\mathbf{j} = -2\mathbf{i} + 20\mathbf{j}$$

A1

3

(b) $\mathbf{r} = (-2\mathbf{i} + 20\mathbf{j}) \times 3 + \frac{1}{2} (2\mathbf{i} - 3\mathbf{j}) \times 9$

Using constant acceleration equation to find $\mathbf{r}$
Correct equation

M1

Correct $\mathbf{r}$

A1

$$3\mathbf{i} + 46.5\mathbf{j}$$

A1

$$r = \sqrt{3^2 + 46.5^2} = 46.6 \text{ ms}^{-1}$$

Finding magnitude
Correct magnitude

M1

5

(c) $\mathbf{v} = (2T - 2)\mathbf{i} + (20 - 3T)\mathbf{j}$
Forming equation for v

M1

$$100 = (2T - 2)^2 + (20 - 3T)^2$$

Forming equation that T must satisfy

M1

$$13T^2 - 128T + 304 = 0$$

Correct Equation

A1

$$T = \frac{128 \pm \sqrt{128^2 - 4 \times 13 \times 304}}{2 \times 13}$$

Solving for T

M1

= 4 or 5.85
Correct solutions

A1

$T = 4$
Selecting $T = 4$

A1

6

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[14]

Q4.

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(a) $\mathbf{a} = \frac{1000t\mathbf{i} - 5000\mathbf{j}}{2000} = \frac{t}{2}\mathbf{i} - \frac{5}{2}\mathbf{j}$
Using Newton's 2nd law

M1

Correct acceleration

A1

2

(b) $\mathbf{v} = \left(\frac{t^2}{4} + c\right)\mathbf{i} + \left(\frac{-5t}{2} + d\right)\mathbf{j}$
Integration

M1

Correct integral with or without constants

A1

$$t = 0, \mathbf{v} = 6\mathbf{j} \Rightarrow c = 0, d = 6$$

Evaluation of constants

m1

Correct constants and correct final answer

A1

4

$$\mathbf{v} = \left(\frac{t^2}{4}\right)\mathbf{i} + \left(6 - \frac{5t}{2}\right)\mathbf{j}$$

$$(c) \quad \mathbf{r} = \left(\frac{t^3}{12} + e\right)\mathbf{i} + \left(6t - \frac{5t^2}{4} + f\right)\mathbf{j}$$

Integration

M1

Correct integral with or without constants

A1

$$t = 0, \mathbf{r} = 0\mathbf{i} + 0\mathbf{j} \Rightarrow e = 0, f = 0$$

Evaluation of constants

m1

$$\mathbf{r} = \left(\frac{t^3}{12}\right)\mathbf{i} + \left(6t - \frac{5t^2}{4}\right)\mathbf{j}$$

Correct constants and correct final answer

A1

4

[10]

Q5.

$$(a) \quad \mathbf{r} = (4\mathbf{i} + 7\mathbf{j})t + \frac{1}{2}(-2\mathbf{i} + 4\mathbf{j})t^2$$

Finding $\mathbf{r}$

M1

$$= (4t + t^2)\mathbf{i} + (7t - 2t^2)\mathbf{j}$$

Correct $\mathbf{r}$

A1

2

(b) $4t - t^2 = 3$

Forming equation for one component

M1

$$t^2 - 4t + 3 = 0$$

$$(t - 1)(t - 3)$$

Solving quadratic equation

m1

$$(t - 1)(t - 3)$$

Solving quadratic equation

m1

$$t = 1 \text{ or } t = 3$$

Two solutions

A1

$$7t - 2t^2 = 5$$

Forming and solving quadratic equation

M1

$$2t^2 - 7t + 5 = 0$$

$$(2t - 5)(t - 1)$$

$$t = 1 \text{ or } t = 2.5$$

Two solutions

A1

$$t = 1$$

Selecting correct solution

A1

6

[8]

Q6.

(a) (i) $\vec{AB} = (70\mathbf{i} - 40\mathbf{j}) - (30\mathbf{i} - 35\mathbf{j}) = 40\mathbf{i} - 5\mathbf{j}$
cao

B1

1

$$(ii) \quad 40\mathbf{i} - 5\mathbf{j} = 10\mathbf{v} + \frac{1}{2} (0.2\mathbf{i} + 0.1\mathbf{j}) \times 100$$

Using their $\vec{AB}$

Correct RHS of the equation

M1 A1

$$10\mathbf{v} = 30\mathbf{i} - 10\mathbf{j}$$

correct $\mathbf{v}$

$$\mathbf{v} = 3\mathbf{i} - \mathbf{j}$$

A1

3

$$(b) \quad 3\mathbf{i} - \mathbf{j} = \mathbf{u} + (0.2\mathbf{i} + 0.1\mathbf{j}) \times 10$$

Correct RHS

M1 A1

$$\mathbf{u} = \mathbf{i} - 2\mathbf{j}$$

Correct $\mathbf{u}$

A1

3

$$(c) \quad 30\mathbf{i} - 35\mathbf{j} = (\mathbf{i} - 2\mathbf{j}) \times 10 + \frac{1}{2} (0.2\mathbf{i} + 0.1\mathbf{j}) \times 100 + \mathbf{r}_0$$

Equation of correct form for $t = 10$ or 20 with their $\mathbf{u}$

Correct equation with their $\mathbf{u}$

M1 A1

$$30\mathbf{i} - 35\mathbf{j} = 20\mathbf{j} - 15\mathbf{j} + \mathbf{r}_0$$

$$\mathbf{r}_0 = 10\mathbf{i} - 20\mathbf{j}$$

Correct position vector

A1

3

[10]

Q7.

$$(a) \quad \mathbf{v} = -2\sin\left(\frac{t}{20}\right)\mathbf{i} + 4\cos\left(\frac{t}{20}\right)\mathbf{j}$$

Differentiating

i component correct

j component correct

M1 A1 A1

3

(b) $\mathbf{v} = 4\mathbf{j}$

Correct velocity at $t = 0$

B1

Travelling north

Travelling north with $\mathbf{v} = n\mathbf{j}$ where $n > 0$

B1

2

(c) $2 \sin\left(\frac{t}{20}\right) = 0$

Equation for t based on $0i$

M1

$$\frac{t}{20} = \pi$$

$$t = 20\pi$$

Correct solution

A1

2

[7]

Q8.

EXAM PAPERS PRACTICE

(a) $\mathbf{F} = (2 + 6 - 2)\mathbf{i} + (3 - 8 + 10)\mathbf{j}$

Summing the forces

M1

$$= 6\mathbf{i} + 5\mathbf{j}$$

Correct resultant

A1

2

(b) $\mathbf{a} = \frac{1}{8} (6\mathbf{i} + 5\mathbf{j}) = 0.75\mathbf{i} + 0.625\mathbf{j}$

Dividing by 8 to get correct value

B1

1

(c) (i) $\mathbf{v} = (3\mathbf{i} - 2\mathbf{j}) + 40(0.75\mathbf{i} + 0.625\mathbf{j})$

use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$

M1

$= 33\mathbf{i} + 23\mathbf{j}$

Correct $\mathbf{v}$

A1

2

(ii) $\mathbf{r} = (3\mathbf{i} - 2\mathbf{j}) \times 40 + \frac{1}{2} (0.75\mathbf{i} + 0.625\mathbf{j}) \times 40^2$

Vector constant acceleration equation to find $\mathbf{r}$

Correct equation

M1 A1

$= 720\mathbf{i} + 420\mathbf{j}$

Correct position vector

A1

$r = \sqrt{720^2 + 420^2} = 834 \text{ m}$

Finding magnitude

Correct magnitude

M1 A1

5

[10]

Q9.

(a) $44\mathbf{i} + 28\mathbf{j} = 10(4\mathbf{i} + 2\mathbf{j}) + 50\mathbf{a}$

Equation to find $\mathbf{a}$ using position vector

Correct equation

M1/A1

$\mathbf{a} = 0.08\mathbf{i} + 0.16\mathbf{j}$

Solving for $\mathbf{a}$

Correct $\mathbf{a}$

M1/A1

4

(b) $\mathbf{r} = (4\mathbf{i} + 2\mathbf{j})t + \frac{1}{2} (0.08\mathbf{i} + 0.16\mathbf{j})t^2$

Equation for $\mathbf{r}$

One component correct

M1/A1

$$= (4t + 0.04t^2)\mathbf{i} + (2t + 0.08t^2)\mathbf{j}$$

Second component correct

A1

3

(c) (i) $96 = 4T + 0.04T^2 \quad 72 = 2T + 0.08T^2$

Forms and solves equation for one component

M1

$$T = 20 \text{ or } -120 \quad T = 20 \text{ or } -45$$

Correct solutions

A1

Hence $T = 20$

Forms and solves equation for other component
Correct solutions plus correct conclusion

M1/A1

4

(d) $\mathbf{v} = (4\mathbf{i} + 2\mathbf{j}) + (0.08\mathbf{i} + 0.16\mathbf{j}) \times 20$

Finding $\mathbf{v}$ at $t = 20$

M1

$$= 5.6\mathbf{i} + 5.2\mathbf{j}$$

Correct $\mathbf{v}$

A1

$$v = \sqrt{5.6^2 + 5.2^2} = 7.64 \text{ ms}^{-1}$$

Finding correct speed

A1

3

[14]